Chapter 65 — Presence & Typing Indicators

> **Part VIII — Real-Time & Connectivity** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 63 (WebSocket), Ch 64 (chat), Ch 44 (lifecycle), Ch 9 (debounce), Ch 3 (SSOT), Ch 28 (privacy). Pairs with Ch 49 (FCM), Ch 68 (network state).
>
> **Reference surface in StreamCart:** the "live" cues around chat (Ch 64) — "online / last seen", "typing…", and "viewers watching" on a live stream — delivered over the WebSocket (Ch 63) as **ephemeral, high-churn, best-effort** signals that must not be confused with durable state.

65.1 Problem Definition and Real-World Context

Presence ("online/away/last seen"), typing indicators ("Alice is typing…"), and live counts (viewers, active users) are **ephemeral real-time signals**: they change constantly, are only meaningful *right now*, and are **not durable data**. Treating them like messages (Ch 64) — persisting every change, delivering reliably, showing stale values — wastes battery/bandwidth, floods the UI, and misleads users. The engineering is about **cheap, debounced, auto-expiring, best-effort** signals with the right privacy controls.

Concretely, teams hit:

- **Treating presence as durable state.** Persisting "online" and showing it after the connection dropped → **stale "online" ghosts**. Presence must **auto-expire** (heartbeat
    - TTL) and be treated as **transient** (not SSOT like messages, Ch 64).
- **Typing-event spam.** Sending a "typing" event per keystroke floods the socket; must be **debounced/throttled** (Ch 9) with a **start/stop + timeout** model (auto-clear if no update).
- **Stale indicators.** A "typing…" that never clears (sender dropped) or an "online" that lingers → must **expire client-side** on a timeout even without an explicit "stopped" event.
- **Battery/bandwidth.** High-churn signals over a mobile radio are costly; **coalesce** counts (sample/debounce), **don't over-send**, and scope to **active** lifecycle (Ch 44).
- **Privacy.** Presence and read/typing receipts reveal behavior; must be **opt-out/ configurable** and **reciprocal** (if you hide yours, you don't see others' — a common product rule), per privacy settings (Ch 28).
- **Scale (fan-out).** "Viewers watching" or presence in a large group is a **fan-out** problem (server) — the client subscribes to a **presence channel** and renders coalesced updates; it never computes global presence itself.

The architecture: model presence/typing/counts as **ephemeral, best-effort** signals over the **WebSocket** (Ch 63), **not** durable SSOT (contrast Ch 64): **heartbeat + TTL** for presence (auto-expire "online"), **debounced start/stop + client-side timeout** for typing (Ch 9), **coalesced/sampled** counts (backpressure, Ch 63); scope signals to **active lifecycle** (Ch 44) and stop sending when backgrounded; render **transient UI state** that **expires locally** if updates stop; and gate on **privacy settings** (opt-out, reciprocal, Ch 28). This chapter is about **doing lightweight real-time cues correctly — the opposite discipline from durable messaging**.

Three practical use cases

1. **Typing indicator (StreamCart chat, Ch 64).** Debounced "typing" start on first keystroke, auto-"stopped" after a pause or send, with a **client-side timeout** so a dropped sender's "typing…" clears.
2. **Online / last seen.** Presence via **heartbeat + TTL**: a user is "online" only while heartbeats arrive; otherwise show "last seen" (coarsened for privacy, Ch 28), respecting opt-out.
3. **Live viewer count.** A coalesced/sampled count on a live stream — the client renders a smoothed number, never every tick (backpressure, Ch 63).

Why this is architectural

Presence/typing/counts are the **ephemeral counterpart** to durable messaging (Ch 64) over the same transport (Ch 63). The decisions — transient-not-durable, heartbeat/TTL, debounce + client-side expiry, coalescing, lifecycle scoping, privacy — define a reusable discipline for *all* "live cue" features and prevent the classic ghosts/spam/battery bugs.

65.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Show presence (online/away/last seen) via heartbeat + TTL (auto-expire).
F2	Show typing indicators (debounced start/stop + client-side timeout).
F3	Show live counts (viewers/active) coalesced/sampled.
F4	Treat all as ephemeral/best-effort (not durable SSOT); expire locally.
F5	Scope signals to active lifecycle; stop when backgrounded (Ch 44).
F6	Respect privacy settings (opt-out, reciprocal) (Ch 28).

Non-functional requirements

Attribute	Policy
Freshness	Reflects now; auto-expires stale (no ghosts).
Efficiency	Debounced/coalesced; low bandwidth/battery; active-only.
Best-effort	Loss is acceptable (unlike messages, Ch 64).
Privacy	Configurable + reciprocal; coarsened last-seen (Ch 28).
Scale	Server fan-out; client renders coalesced updates.

Scope

- **In scope:** presence (heartbeat + TTL, online/away/last-seen), typing (debounce + start/stop + client-side timeout, Ch 9), live counts (coalesce/sample, Ch 63), ephemeral transient state (not SSOT), lifecycle scoping (Ch 44), privacy (opt-out, reciprocal, coarsened, Ch 28), presence channels / fan-out (client side).
- **Out of scope:** WebSocket transport (Ch 63), chat/messages (Ch 64), FCM (Ch 49), debounce mechanics (Ch 9), network state (Ch 68), privacy framework (Ch 28), server fan-out implementation.

Assumptions & constraints

- WebSocket (Ch 63) as transport; signals are **ephemeral** (lost signals are fine); server does presence **fan-out** and may compute aggregate presence/counts; `minSdk 24`. Presence/typing are **not** persisted as durable truth (contrast Ch 64); last-seen may be persisted coarsely server-side.

65.3 System Design Fundamentals

- **Ephemeral, best-effort — the opposite of messages (Ch 64).** Presence/typing/counts are only meaningful **now**; a lost update is **fine** (the next one corrects it). So — unlike chat (Ch 64) — **no outbox, no idempotency-for-durability, no persistent SSOT of truth, no catch-up.** They live as **transient UI state** that **expires** if updates stop. Conflating these with durable messaging is the root mistake.
- **Presence = heartbeat + TTL (auto-expire).** A client that's online sends periodic **heartbeats** (or the WebSocket's liveness, Ch 63, serves as one); the server marks the user **online** and sets a **TTL**. If no heartbeat arrives within the TTL (app closed, network dropped), the user **auto-expires to offline** — no explicit "I went offline" needed (which often never arrives). The client shows "online" only within a fresh TTL; otherwise **"last seen"** (coarsened, Ch 28). This prevents **stale "online" ghosts**.
- **Typing = debounced start/stop + client-side timeout (Ch 9):**
  - **Start:** on the first keystroke, send a **"typing start"** (then **debounce** — don't send again for a few seconds, Ch 9).
  - **Stop:** send **"typing stop"** on send, on clearing the field, or after an **idle timeout** (e.g., 3–5s of no typing).
  - **Client-side expiry (critical):** the *receiver* also runs a **timeout** — if no further "typing" within N seconds, **clear** the indicator anyway. This guarantees a "typing…" clears even if the "stop" event is lost or the sender dropped (best-effort). So typing is **two throttled events + a receiver-side auto-clear**, not a stream.
- **Live counts = coalesce/sample (backpressure, Ch 63).** A rapidly-changing count (viewers) should be **conflated/sampled** so the UI updates at most a few times/second (Ch 63 backpressure) — render the **latest**, not every tick. The server typically sends **aggregate** counts (it does the fan-out/counting); the client just smooths rendering.
- **Transient UI state that expires locally.** Presence/typing/counts are held in **in-memory UI state** (a `StateFlow`, Ch 6), **not** persisted as truth. Each has an **expiry**: presence TTL, typing timeout, count staleness. On expiry (or disconnect), the UI **falls back** (presence → last-seen/unknown, typing → cleared, count → last-known or hidden). On reconnect (Ch 63), fresh signals repopulate.
- **Lifecycle scoping (Ch 44).** Send presence/typing only while the relevant screen is **active**; **stop** when backgrounded (the socket disconnects anyway, Ch 63) — the server's TTL then expires the user to offline. Don't send typing for a backgrounded chat. This is the main battery/bandwidth lever.

- **Privacy (opt-out + reciprocal, Ch 28).** Presence and typing/read receipts (Ch 64) **reveal behavior**. Make them **configurable**: a user can **hide** their online status/ last-seen/typing; a common rule is **reciprocity** — if you hide yours, you **don't see** others' (symmetry). **Coarsen "last seen"** (e.g., "today", "this week") for privacy. Respect the setting **both ways** and server-enforce it (Ch 28).
- **Scale / fan-out (server).** In a large group or stream, presence/counts are a **fan-out** problem solved server-side (presence service, pub/sub). The client **subscribes** to a presence/count **channel** (authorized, Ch 63/23) and renders coalesced updates — it never computes global presence itself.

Core insight: **presence, typing, and live counts are ephemeral, best-effort real-time signals — the opposite discipline from durable messaging (Ch 64): no persistence/outbox/ idempotency-for-truth, just transient UI state that auto-expires. Use heartbeat + TTL so presence auto-expires to offline (no ghosts), debounced start/stop plus a receiver-side timeout so typing clears even when the stop event is lost, and coalescing/sampling for high-churn counts (Ch 63); scope sending to the active lifecycle (stop when backgrounded, Ch 44) as the main battery lever; and gate everything on privacy settings (opt-out, reciprocal, coarsened last-seen, Ch 28), with the server doing fan-out.**

65.4 High-Level Architecture


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Components & ownership

- **Presence:** heartbeat (active) → server TTL → auto-expire; client subscribes.
- **Typing:** debounced start/stop → server fan-out → receiver auto-clears on timeout.
- **Counts:** server aggregate → client coalesces/samples.
- **All → transient UI state** that expires locally; **privacy** gates send/receive; **lifecycle** stops sending in background.

Direction: active signals → server (TTL/fan-out) → subscribe → coalesced transient UI state → local expiry; privacy + lifecycle gate it.

65.5 Detailed Component Design


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Component	Responsibility
`PresenceTracker`	heartbeat (active); observe TTL-expiring presence
`TypingController`	debounced start/stop; receiver-side timeout auto-clear (Ch 9)
`LiveCount`	coalesced/sampled aggregate count (Ch 63)
`PrivacySettings`	opt-out + reciprocal gating (Ch 28)

65.6 End-to-End Data Flow

Typing: debounce + receiver timeout


Typing: debounce + receiver timeout

Presence: heartbeat + TTL auto-expire


Presence: heartbeat + TTL auto-expire

65.7 Complete Kotlin Implementation

Package: `com.streamcart.presence`.

65.7.1 Presence: heartbeat (active) + TTL-expiring observation

```
@Singleton
class PresenceTracker @Inject constructor(
    private val transport: RealtimeTransport,     // WebSocket (Ch 63)
    private val privacy: PrivacySettings, private val scope: CoroutineScope,
) {
    private var heartbeatJob: Job? = null
    /** Send heartbeats only while active (Ch 44); server TTL auto-expires us when we stop. */
    fun startHeartbeat() {
        if (!privacy.sharePresence()) return                    // opt-out (Ch 28)
        heartbeatJob = scope.launch {
            while (isActive) { transport.send(Presence.Heartbeat); delay(HEARTBEAT_MS) }
        }
    }
    fun stopHeartbeat() { heartbeatJob?.cancel() }              // background -> stop -> server
TTL expires us

    /** Observe another user's presence; render online only within a fresh TTL, else last-seen.
*/
    fun observe(userId: String): Flow<Presence> = transport.presenceUpdates(userId)
        .filter { privacy.canSeePresence() }                    // reciprocal (Ch 28)
        .map { update -> if (update.ageMs < TTL_MS) Presence.Online else
Presence.LastSeen(update.lastSeenCoarse) }
}
```

65.7.2 Typing: debounced start/stop (Ch 9)

```
class TypingController @Inject constructor(
    private val transport: RealtimeTransport, private val privacy: PrivacySettings,
) {
    private val input = MutableSharedFlow<Unit>(extraBufferCapacity = 64)
    fun bind(conversationId: String, scope: CoroutineScope) {
        if (!privacy.shareTyping()) return
        scope.launch {
            input
                .onEach { transport.send(Typing.Start(conversationId)) }   // start on first
input
                .debounce(TYPING_DEBOUNCE_MS)                              // throttle (Ch 9) -
don't spam per keystroke
                .collect { }                                               // (start already
sent; debounce gates frequency)
        }
        scope.launch {
            input.debounce(TYPING_IDLE_MS)                                 // idle -> stop (Ch
9)
                .collect { transport.send(Typing.Stop(conversationId)) }
        }
    }
    fun onInput() { input.tryEmit(Unit) }
    fun onSendOrClear(conversationId: String) { transport.send(Typing.Stop(conversationId)) }
// explicit stop
}
```

65.7.3 Receiver-side typing with client timeout (auto-clear)

```
class TypingReceiver @Inject constructor(private val transport: RealtimeTransport) {
    /** Clear a user's "typing" if no update within the timeout — even if "stop" was lost. */
    fun observeTyping(conversationId: String): Flow<Set<String>> = channelFlow {
        val typing = mutableMapOf<String, Job>()
        val state = MutableStateFlow<Set<String>>(emptySet())
        launch { state.collect { send(it) } }
        transport.typingUpdates(conversationId).collect { ev ->
            when (ev) {
                is Typing.Start -> {
                    typing[ev.userId]?.cancel()
                    typing[ev.userId] = launch {                        // client-side expiry
(best-effort)
                        delay(TYPING_RECEIVER_TIMEOUT_MS); state.update { it - ev.userId }  //
auto-clear
                    }
                    state.update { it + ev.userId }
                }
                is Typing.Stop -> { typing.remove(ev.userId)?.cancel(); state.update { it -
ev.userId } }
            }
        }
    }
}
```

65.7.4 Live count: coalesce/sample (backpressure, Ch 63)

```
fun RealtimeTransport.viewerCount(channel: String): Flow<Int> =
    countUpdates(channel)                // server sends aggregate
        .conflate()                      // keep latest only
        .sample(500.milliseconds)        // at most ~2 updates/sec (Ch 63 backpressure)
// Transient: on disconnect, show last-known or hide; repopulates on reconnect.
```

65.7.5 Lifecycle scoping + transient UI state (not SSOT)

```
// Presence/typing are transient StateFlow (Ch 6), NOT persisted as truth (contrast Ch 64).
@Composable
fun ChatHeader(userId: String, vm: ChatViewModel = hiltViewModel()) {
    val presence by vm.presence(userId).collectAsStateWithLifecycle(Presence.Unknown)  //
expires to LastSeen/Unknown
    val typing by vm.typing.collectAsStateWithLifecycle(emptySet())                     // auto-
clears on timeout
    // Start/stop heartbeat with lifecycle (Ch 44): onStart -> startHeartbeat(); onStop ->
stopHeartbeat()
    Text(when (presence) { Presence.Online -> "Online"; is Presence.LastSeen -> "Last seen
${presence.text}"; else -> "" })
    if (typing.isNotEmpty()) Text("typing…")
}
```

65.7.6 Module structure

```
:core:presence  PresenceTracker, TypingController, TypingReceiver, LiveCount, PrivacySettings
# Transport -> :core:realtime WebSocket (Ch 63). Lifecycle -> Ch 44. Privacy -> Ch 28.
# NOT persisted as SSOT (contrast chat messages, Ch 64).
```

65.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Presence "online" ghost after drop	stale UI	heartbeat + TTL auto-expire (server + client age check)
Typing "…" never clears	stuck UI	receiver-side timeout auto-clear (best-effort)
Typing-event spam	flood/battery	debounce start + idle-stop (Ch 9)
Count flood	UI jank	conflate + sample (Ch 63)
Lost stop/heartbeat	stale	local expiry handles it (don't rely on explicit stop)
Sending while backgrounded	battery	stop on background; socket disconnects (Ch 44/63)
Privacy mismatch	leak	reciprocal gate both ways; server-enforced (Ch 28)

- **Local expiry is the key safety net:** because signals are best-effort, the **receiver** (not just the sender) enforces expiry — presence ages out via TTL, typing clears on a timer. This means a **lost "stop"/heartbeat never leaves a stuck indicator** (the classic ghost/stuck-typing bug).
- **Debounce prevents spam without losing meaning:** a "typing start" + idle "stop" (Ch 9) conveys the state in 2 events instead of one-per-keystroke, saving radio/battery.
- **Conflation handles high-churn counts:** render the latest sampled value; dropping intermediate counts is correct for an ephemeral signal.

65.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Stale "online"	user report	heartbeat + TTL (no explicit offline needed)	accurate
Stuck "typing…"	user report	receiver timeout auto-clear	clears
Event spam/battery	metrics	debounce/throttle (Ch 9)	efficient
Count flicker/flood	jank	conflate + sample (Ch 63)	smooth
Signals sent in background	battery	lifecycle-scoped; stop (Ch 44)	battery ok
Presence leaks despite opt-out	privacy audit	reciprocal + server-enforced (Ch 28)	private
Disconnect clears cues	UX	show last-known/last-seen; repopulate on reconnect	graceful


65.9 Failure Scenarios and Recovery

> **The whole discipline is "ephemeral, auto-expiring, best-effort" — the inverse of chat (Ch 64).** No persistence, no delivery guarantees; instead heartbeat+TTL (presence), debounce + receiver-timeout (typing), and conflate+sample (counts), all scoped to the active lifecycle and gated by privacy. Get expiry right and ghosts/stuck-indicators/spam all disappear.

65.10 Scaling the Implementation

- **Level 1.** Typing: debounced start/stop + receiver timeout; presence: online/offline via WebSocket liveness.
- **Level 2.** Presence heartbeat + TTL + coarsened last-seen; live counts coalesced (Ch 63); lifecycle scoping (Ch 44).
- **Level 3 — production.** Privacy (opt-out + reciprocal, Ch 28); transient UI state with local expiry; presence channels (authorized subscribe, Ch 63/23).
- **Level 4 — large.** Server-side presence fan-out/pub-sub at scale; group presence (member lists); aggregate counts; per-feature coalescing policies.
- **Level 5 — enterprise.** Presence-service SLOs, privacy governance (regulatory), cross-platform consistency, presence analytics (engagement), abuse protection.

Avoid overengineering: typing (debounce + timeout) and presence (heartbeat + TTL) are small; don't persist them or build delivery guarantees. Reuse the same transient/expiry pattern for all live cues.

65.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Model	Ephemeral transient (expire locally)	durable SSOT like messages (Ch 64)	Ephemeral always (presence/typing/counts)
Presence	Heartbeat + TTL (auto-expire)	explicit online/offline events	Heartbeat+TTL (offline event often lost)
Typing	Debounce start/stop + receiver timeout	per-keystroke stream	Debounce+timeout (spam/battery)
Counts	Coalesce + sample (Ch 63)	render every tick	Coalesce (UI/battery)
Expiry	Receiver-side timeout	trust explicit stop	Receiver timeout (best-effort; stop can be lost)
Sending	Active lifecycle only	always	Active-only (battery, Ch 44)
Privacy	Opt-out + reciprocal (Ch 28)	always share	Opt-out+reciprocal (trust)

Ephemeral vs durable modeling: the biggest mistake is treating presence/typing like **messages** (Ch 64) — persisting them, guaranteeing delivery, showing last value. They're **ephemeral**: meaningful only now, cheap to lose, must **auto-expire**. Modeling them as **transient UI state with local expiry** (not SSOT, no outbox/idempotency-for-truth) is simpler, cheaper, and correct; durable modeling causes ghosts, spam, and wasted storage.

Heartbeat+TTL vs explicit online/offline events: sending an explicit "I'm offline" when the app closes is unreliable — the app may be killed, the network dropped, before it can send. **Heartbeat + server TTL** needs **no** reliable offline event: absence of heartbeats expires the user. Always use heartbeat+TTL for presence; never depend on a graceful offline signal.

Receiver-side timeout vs trusting "stop": if a "typing stop" (or a user drop) is lost, trusting the explicit stop leaves a **stuck "typing…"**. A **receiver-side timeout** that clears after N seconds of silence guarantees the indicator clears regardless — the standard best-effort pattern for all ephemeral cues.

65.12 Security & Privacy Considerations

- **Presence/typing/read receipts reveal behavior (Ch 28):** when you're online, when you read/typed. Make them **opt-out** and respect the setting; **server-enforce** it (a client can't be trusted to hide another user's view).
- **Reciprocity:** a common, user-trusted rule — if you **hide** your presence/receipts, you **don't see** others' (symmetry). Implement both directions.
- **Coarsen last-seen (Ch 28):** show "today"/"this week" rather than an exact timestamp to reduce behavioral tracking; let users hide it entirely.
- **Authorized channels (Ch 63/23):** subscribing to a presence/typing channel for a conversation/stream must be **server-authorized** (membership, Ch 64/23) — you can't watch arbitrary users' presence.
- **Per-account isolation (Ch 42):** presence/typing scoped to the active account; stop on switch/logout.
- **No presence/behavior data in logs** (Ch 28); aggregate counts only.
- Client vs backend: the **server** enforces privacy settings, authorizes channels, does fan-out, and computes aggregate presence/counts; the **client** gates send/display by settings and renders transient, expiring state — it

never computes or exposes others' presence against their settings.

65.13 Performance and Resource Optimization

- **Lifecycle scoping (Ch 44)** — send presence/typing only while active; stop on background (the top battery lever; the socket disconnects anyway, Ch 63).
- **Debounce/throttle** typing (Ch 9) — 2 events, not one-per-keystroke.
- **Heartbeat interval** tuned — frequent enough for accurate presence, infrequent enough to save radio (align with socket keep-alive, Ch 63).
- **Coalesce/sample counts** (Ch 63) — render latest; drop intermediate ticks.
- **No persistence** — transient in-memory state; no DB writes for high-churn signals.
- **Batch presence for lists** — one subscription per conversation list, aggregate render.
- **Profiling:** presence/typing event rate, battery/radio from signals, count update frequency, ghost/stuck-indicator incidents (→ 0), subscription count.

65.14 Testing Strategy

Expiry/debounce logic is very testable with virtual time; transport faked.

```
class TypingReceiverTest {
    @Test fun `clears typing after timeout even if stop is lost`() = runTest {
        val flow = receiver.observeTyping("c1")
        transport.emit(Typing.Start("alice"))
        advanceTimeBy(TYPING_RECEIVER_TIMEOUT_MS + 1)           // no stop sent
        assertEquals(emptySet<String>(), flow.first { it.isEmpty() })   // auto-cleared (best-
effort)
    }
    @Test fun `debounces typing to start + idle-stop, not per keystroke`() = runTest {
        repeat(10) { controller.onInput(); advanceTimeBy(100) } // rapid typing
        assertEquals(1, transport.sent.count { it is Typing.Start })    // one start (Ch 9)
        advanceTimeBy(TYPING_IDLE_MS); assertTrue(transport.sent.any { it is Typing.Stop })  //
idle stop
    }
}

class PresenceTest {
    @Test fun `shows last-seen when heartbeat older than TTL`() = runTest {
        val p = tracker.observe("bob").first()                 // update age > TTL
        assertTrue(p is Presence.LastSeen)                      // no ghost
    }
    @Test fun `respects presence opt-out and reciprocity`() { /* sharePresence=false -> no
heartbeat, canSee=false */ }
}

class LiveCountTest { @Test fun `samples rapid counts`() = runTest { /* 100 ticks -> few
emissions */ } }
```

What to test: typing debounce (start + idle-stop, Ch 9) + receiver timeout auto-clear (lost-stop case); presence TTL → last-seen (no ghost); privacy opt-out + reciprocity; count conflate/sample; lifecycle stop on background (no sends). Use `runTest` virtual time; fake transport.

65.15 Production Observability

- **Metrics:** presence/typing **event rate** (and battery/radio attributed); ghost- presence + stuck-typing incidents (→ 0); count update frequency; privacy opt-out rate; subscription count; background-send incidents (should be 0, Ch 44).
- **Logs:** `presence{event, expired}` / `typing{event, source}` — **no per-user behavior/ PII** (Ch 28); aggregates only.
- **Debugging:** stale "online" → TTL/heartbeat (server or client age check); stuck "typing" → receiver timeout missing; spam/battery → no debounce / not lifecycle-scoped; count jank → no coalescing; privacy leak → reciprocity/server enforcement (Ch 28).
- **Alerting:** signal-event-rate spike (spam regression); battery attributed to presence; any privacy-setting bypass.

- **Dashboards:** signal rate + battery; ghost/stuck incident trend.

65.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Presence/typing/counts vs messages (Ch 64)?* Ephemeral/best-effort vs durable.
2. *Persist them?* No — transient UI state that expires.
3. *Why best-effort?* Only meaningful now; loss is fine (next update corrects).
4. *Transport?* WebSocket (Ch 63).
5. *Biggest mistake?* Treating them as durable (ghosts/spam).

Presence 6. *Model?* Heartbeat + TTL (auto-expire). 7. *Why not explicit offline event?* Often lost (app killed/network drop). 8. *Show online when?* Within a fresh TTL; else last-seen. 9. *Last-seen privacy?* Coarsen (today/this week); hideable (Ch 28).

Typing 10. *Model?* Debounced start + idle/explicit stop + receiver timeout. 11. *Why debounce?* Per-keystroke floods the socket (Ch 9). 12. *Why receiver timeout?* Stop event / sender can be lost → clear anyway. 13. *When stop?* Send, clear field, idle timeout.

Counts 14. *High-churn counts?* Coalesce + sample (Ch 63). 15. *Who computes them?* Server aggregate; client smooths.

Lifecycle/efficiency 16. *When to send?* Active only; stop on background (Ch 44). 17. *Main battery lever?* Lifecycle scoping + debounce. 18. *Heartbeat interval?* Tuned vs accuracy/battery.

Privacy 19. *Why privacy-sensitive?* Reveals behavior. 20. *Opt-out?* Yes, both ways; server-enforced (Ch 28). 21. *Reciprocity?* Hide yours → don't see others'. 22. *Channel authz?* Server-authorized subscription (Ch 23).

Testing/Production 23. *Test expiry?* Virtual time: lost-stop clears; TTL → last-seen. 24. *Ghost "online"?* TTL/heartbeat bug. 25. *Stuck "typing"?* Missing receiver timeout. 26. *Spam/battery?* No debounce / not lifecycle-scoped.

Design-change challenges

1. **"Online" lingers after users leave.** Heartbeat + TTL auto-expire.
2. **"Typing…" gets stuck.** Receiver-side timeout auto-clear.
3. **Typing floods the socket.** Debounce start + idle-stop (Ch 9).
4. **Viewer count flickers.** Conflate + sample (Ch 63).
5. **Let users hide online status.** Opt-out + reciprocal, server-enforced (Ch 28).

Mock walkthrough (abridged)

Clarify (presence/typing/counts, scale, privacy?) → ephemeral/best-effort (not SSOT, contrast Ch 64) → presence heartbeat+TTL → typing debounce start/stop + receiver timeout (Ch 9) → counts coalesce/sample (Ch 63) → transient UI state with local expiry → lifecycle scoping (Ch 44) → privacy opt-out+reciprocal (Ch 28) → server fan-out → observability (event rate, ghosts).

65.17 Practical Exercises

1. **Beginner.** Add a typing indicator: debounced start, idle/explicit stop, and a receiver-side timeout that clears even if "stop" is lost. *Eval:* no spam; never stuck.
2. **Intermediate.** Implement presence via heartbeat + TTL showing online within a fresh TTL, else coarsened last-seen. *Eval:* no ghost; privacy-coarsened.
3. **Advanced.** Add a coalesced/sampled live viewer count and scope all signals to the active lifecycle (stop on background). *Eval:* smooth count; no background sends.
4. **Production debugging.** Fix a stuck "typing…" (no receiver timeout) and a stale "online" (no TTL). *Eval:* both auto-expire.
5. **System design challenge.** Add opt-out + reciprocal privacy (server-enforced) for presence/typing. *Eval:* hidden both ways; can't see others' when hidden.

(Solutions withheld unless requested.)

65.18 Chapter Summary & Reference Sheet

Key decisions: treat presence, typing, and live counts as **ephemeral, best-effort real-time signals** over the **WebSocket** (Ch 63) — the **opposite discipline from durable messaging** (Ch 64): **no persistence, no outbox, no idempotency-for-truth, no catch-up**, just **transient UI state that auto-expires**. Do **presence** with **heartbeat

- server TTL** so a user **auto-expires to offline** (no stale "online" ghosts — never depend on an explicit offline event); do **typing** with a **debounced start + idle/ explicit stop** (Ch 9) **plus a receiver-side timeout** that clears the indicator even when the "stop" is lost; do **live counts** with **coalescing/sampling** (Ch 63 backpressure, rendering the server's aggregate). **Scope sending to the active lifecycle** and **stop on background** (Ch 44) as the main battery lever; render **transient state with local expiry** (presence → last-seen, typing → cleared, count → last-known); and gate everything on **privacy** — **opt-out**, **reciprocal** (hide yours → don't see others'), **coarsened last-seen**, **server-enforced** (Ch 28) — with the **server doing fan-out** and the client subscribing to authorized channels (Ch 23).

Concepts: ephemeral vs durable (contrast Ch 64), heartbeat + TTL presence, debounced typing + receiver timeout, coalesced counts, transient UI state with local expiry, lifecycle scoping, privacy (opt-out/reciprocal/coarsened), server fan-out.

Common mistakes: modeling presence/typing as durable (ghosts, spam, wasted storage); relying on explicit offline/stop events (lost → stuck); per-keystroke typing (flood); rendering every count tick (jank); sending while

backgrounded (battery); no local expiry (stuck indicators); ignoring privacy/reciprocity; computing global presence client-side.

Checklist: ☐ ephemeral/transient (not SSOT) ☐ presence: heartbeat + TTL auto-expire ☐ typing: debounce start/stop (Ch 9) + receiver timeout auto-clear ☐ counts: conflate + sample (Ch 63) ☐ local expiry (presence/typing/count) ☐ lifecycle-scoped send; stop on background (Ch 44) ☐ privacy opt-out + reciprocal + coarsened + server-enforced (Ch 28) ☐ authorized channels (Ch 23) ☐ per-account (Ch 42) ☐ no persistence/PII logging ☐ event-rate

- ghost/stuck telemetry.

Key APIs: WebSocket transport (Ch 63), `Flow.debounce` / `sample` / `conflate` (Ch 9/63), `MutableStateFlow` / `channelFlow` (transient state, Ch 6), coroutine timeouts (`delay` / cancel for receiver expiry), `ProcessLifecycleOwner` /lifecycle (Ch 44), privacy settings (Ch 28), `collectAsStateWithLifecycle`.

References:

- Flow operators (debounce/sample/conflate): https://kotlinlang.org/api/kotlinx.coroutines/kotlinx-coroutines-core/kotlinx.coroutines.flow/
- (composes) Ch 63 WebSocket, Ch 64 Chat, Ch 9 Debounce, Ch 44 Lifecycle, Ch 28 Privacy, Ch 6 UI State

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/flow/failure diagrams; presence heartbeat+TTL + typing debounce + receiver-timeout + count coalesce + privacy code; concurrency (local expiry safety net, debounce, conflation); failure (ephemeral- discipline principle); 5 levels; ephemeral-vs-durable, heartbeat+TTL, receiver-timeout trade-offs; security/privacy (opt-out, reciprocal, coarsened, channel authz); perf (lifecycle/debounce/coalesce/no-persist); virtual-time tests; observability (event rate, ghosts); 26 Q&A + 5 challenges. ✅
- **Expand later:** WebSocket (Ch 63), chat (Ch 64), privacy (Ch 28), network state (Ch 68).
- **Interview coverage:** exceeds ≥25 bar.